MILITARY SPECIFICATION

BLOOD COLLECTING-DISPENSING BAG AND DONOR SET, DOUBLE BAG, WITH 63 cc ANTICOAGULANT, DISPOSABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a sterile plastic blood collecting and dispensing bag, with integral donor set and secondary bag, suitable for collection, storage, component separation, and administration of blood. Potency period is 48 months.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-636	Box, Shipping, Fiberboard.
PPP-C-96	Cans, Metal, 28 Gage and Lighter.

Military

MIL-B-36939	Blood, Collecting-Dispensing Bag and Donor Set, with Anticoagulant (CPD)

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. PHARMACOPEIAL CONVENTION, INC.
Pharmacopeia of the United States (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042).

U. S. DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Food and Drug Administration - Federal Food, Drug, and Cosmetic Act and Pertinent Regulations.

An approved New Drug Application must be on file with the Food and Drug Administration. Information relative to such applications and to the Federal Food, Drug, and Cosmetic Act may be obtained from:

Bureau of Medicine
Food and Drug Administration
U. S. Department of Health, Education, and Welfare
Washington, D. C. 20204

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE
Bureau of Biologics, Food and Drug Administration
Code of Federal Regulation Part 273
Minimum Requirements (as applicable).

(Application for copies should be addressed to the Food and Drug Administration, 5600 Fishers Lane, Rockville, Maryland 20852.)

3. REQUIREMENTS

3.1 Material. Shall be in accordance with requirements for materials contained in MIL-B-36939 Blood Collecting-Dispensing Bag and Donor Set, with Anticoagulant (CPD), 450 cc, Disposable, 6s.

3.2 Style, design and construction. The set shall consist of a sterile, pyrogen free, nontoxic, disposable, plastic, blood collecting, component processing and dispensing system of two connected plastic bags (i.e., a primary and secondary bag), with anticoagulant solution, integral donor set (with integrally connected phlebotomy needle and donor tube, connecting tube, and control clamp as one unit. The needle, donor tube, primary bag, connecting tube, and secondary bag shall be integrally connected. The set shall be suitable for collecting, storing, and processing 450 milliliters (ml.) of blood into its components within a completely closed and tamper-proof system. Design of the unit shall permit the collection of pilot samples of unmistakable identity for grouping and typing, crossmatching and culturing without entering the closed system of the primary or secondary bags. The unit shall be in conformance with all requirements of the Federal Food, Drug, and Cosmetic Act, and the Pharmacopeia of the United States, as applicable. The assembled unit shall contain 63 cc of anticoagulant solution.

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3.2.1 Primary bag. Except as specified herein, the primary bag shall be 450 ml. blood collecting-dispensing bag in accordance with MIL-B-36939. The arrangement of the donor tube, connecting tube, and two outlet ports, as they are attached to the primary bag, shall be such that when the primary bag is in its upright position these four connections shall be at the top of the primary bag. They shall be arranged so that the connecting tube and donor tube junctions are between and higher than the outlet ports. Their arrangement shall be such as to provide maximum efficiency for component separation of blood through the connecting tube. The outlet ports shall consist of externally protruding tubes hermetically sealed to the bag. The entire exterior surface of the protruding tube shall be sterile and have its sterility maintained by a tamper-proof protective sleeve or rubber plug. The outlet port design shall be such that the piercing device of the blood recipient set will make a leakproof fluid path connection with the outlet port. The ports shall be so constructed as to protect the walls of the bag against piercing by the recipient set and shall provide a proper, leakproof fit with the blood recipient set. The bag shall have slits (or openings) in the side seals for blood collecting tubes.

3.2.1.1 Donor tube and needle. The interior lumen of the connection between the needle cannula and the donor tube shall present smoothly expanding surfaces to minimize the formation of eddy currents, to the extent of preventing platelet aggregation and clotting. All interior and exterior surfaces of the needle cannula which come into contact with blood shall be hemo repellent to the extent of preventing platelet aggregation and clotting. Needle shall be protected by a suitable sheath or sleeve which shall make an interference sealed fit with an internal surface of the needle hub in such a manner that only sterile surfaces and edges of the sheath can contact the point of the needle cannula during removal of the sheath or sleeve. Needle may be treated with silicone, in addition not more than 1.0 mg of total residue per needle shall be present when determined by an appropriate, accurate, and reproducible method. The needle shall be free from foreign matter upon normal visual examination after the equipment has been sterilized.

3.2.2 Secondary bag (transfer bag). Except as hereinafter specified, the secondary bag shall be constructed in the same manner as the primary bag (see 3.2.1). The secondary bag shall comply with Material, Bag strength, Filling and infusion, Bag suspension, In vivo survival of red blood cells, and Adhesion of label to bags, requirements of MIL-B-36939 , except capacity shall be 300 ml. The inside overall dimensions of the secondary bag shall be 5 $\pm \frac{1}{4}$ inches long by 4-3/4 $\pm$ 1/8 inches wide, when measured flat. The capacity of the bag shall be 300 ml. The secondary bag and connecting tube shall not contain anticoagulant solution of the primary bag. The bag shall be provided with two outlet ports as specified for the primary bag, except shall be positioned as follows: The arrangement of the two outlet ports and the connecting tube, as they are attached to the secondary bag, shall be such that when the

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secondary bag is in its upright position, these three connections shall be at the top of the secondary bag. They shall be so arranged that one of the outlet ports is between and higher than the connecting tube junction and the other outlet port junction or the connecting tube junction is between the two outlet port junctions. Their arrangement shall provide maximum efficiency for component separation of the blood.

3.2.3 Connecting tube. The connecting tube shall consist of a minimum length of 15 inches of tubing having a maximum outside diameter of 3/16 inch. One end of the connecting tube shall be connected to the primary bag and its other end to the secondary bag. The junction of the connecting tube to the primary bag shall possess a device which shall prevent the flow of fluid from the primary bag to the secondary bag until desired. This device shall be capable of allowing the free flow of fluid during the processing of the blood. The connecting tube of each bag shall bear the same identical numbers and marks (x's or *'s) as the donor tube indelibly imprinted along the tubing for the purpose of dividing the connecting tube into segments for pilot samples. The center-to-center distance of these marks shall be 2-1/2 to 3 inches. The connecting tube of each unit shall possess a different number.

3.2.3.1 Security of attachment. The connecting tube shall not become detached, or show any evidence of separation, from the primary or secondary bag, when tested as specified in 4.4.1.

3.2.4 Control clamp. The control clamp for the secondary bag shall provide a constant setting and quick shut-off from the operating position, shall be movable, and shall be capable of one-hand operation. The control clamp shall be either the screw type or the roller type and shall be fabricated of suitable plastic. The design shall be such that the screw or roller cannot be accidentally removed. To assure constant delivery of fluid, it shall be impossible for the clamp setting to be changed by an accidental jerk on the tubing. The control clamp shall be attached in the open position, and the clamp shall not cut or damage the tubing in either the open or closed position. When tested as specified in 4.4.2, the clamp shall be capable of accomplishing complete shut-off, shall regulate flow of the fluid from complete shut-off to maximum flow, and with a constant head of water shall maintain a constant flow rate at any point between complete shut-off and maximum flow. In addition, when the clamp is operated from its fully open position to complete shut-off, and this operation is repeated ten times while the clamp is in the same position on the tubing, the tubing shall not be cut, pierced or damaged so as to cause leakage or restriction of flow of fluid. As an alternate the control clamp shall be a slide clamp.

3.2.5 Anticoagulant solution. The primary bag shall contain 63 cc of sterile, pyrogen free anticoagulant solution. The anticoagulant solution shall be as specified in MIL-B-36939.

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3.3 Sterility. When tested as specified in 4.4.3, the equipment shall comply with sterility test requirements of the Pharmacopeia of the United States (USP).

3.4 Safety, pyrogenicity, and red blood cell survival. The material used to fabricate the equipment shall be of such hydrolytic resistance, insofar as fluids used in blood transfusion are concerned, that no harmful quantities of material having harmful effects are extracted in normal use.

3.4.1 Safety. When tested as specified in 4.4.4, the equipment shall comply with safety test requirements of the USP.

3.4.2 Freedom and pyrogens. When tested as specified in 4.4.5, the equipment shall comply with pyrogen test requirements of the USP.

3.4.3 In vivo survival of red blood cells. The absences of demonstrable detrimental effect on the red cells shall be established by in vivo evidence. The viability of red cells of whole blood collected in the equipment shall be such that after 21 days of storage at $5^{\circ} \pm 1^{\circ}$ C., not less than 70 percent of the red cells are found to be viable, by in vivo testing as specified in 4.4.6. No change in formulation or fabrication of the plastic shall be permitted without submission of evidence that it does not decrease in vivo survival of red cells to less than the level specified above.

3.5 Manufacturer's record. The manufacturer shall place a code or lot number on the paper label, or on the unit package, which can be verified against the manufacturer's records of all tests made on sterility, safety, pyrogenicity and red blood cell survival. These records shall be made available for inspection by any duly qualified medical officer of the Armed Forces, National Institutes of Health, or any official of the Federal Food and Drug Administration. No shipment shall be made until all the tests have been completed, verified and recorded in the permanent records of the manufacturer.

3.6 Certificates. For destination inspection only, certificates of sterility, safety, freedom from pyrogens, and in vivo survival of red blood cells as specified below, shall be submitted by the contractor directly to Technical Operations Division, Directorate of Medical Materiel, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa. 10101, Attention: Quality Assurance Branch, and one copy of each shall be forwarded with each shipment to the consignee. For source inspection certificates shall be furnished to the procuring activity via the government inspector.

3.6.1 Certificate of sterility. A certificate of sterility, listing all sterility lot or control numbers, shall state that samples from each sterility lot or control have been tested for sterility, as required in 3.3, and were found to be sterile. The certificate shall reference the method used in determining sterility.

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3.6.2 Certificate of safety. A certificate of safety, listing all lot or control numbers, shall state that samples from each lot or control have been tested for safety as specified in 3.4.1, and were found to comply with such requirements. The certificate shall reference the method used in determining safety.

3.6.3 Certificate of freedom from pyrogens. A certificate of freedom from pyrogens, listing all lot or control numbers, shall state that samples from each lot or control have been tested for pyrogens as specified in 3.4.2 and were found to comply with such requirements. The certificate shall reference the method used in determining freedom from pyrogens.

3.6.4 Certificate of in vivo survival of red blood cells. A certificate of in vivo survival of red blood cells, listing lot or batch numbers, shall state that samples from each lot or batch of plastic have been tested for red blood cell survival as specified in 3.4.3 and were found to comply with such requirements. The certificate shall reference the method used in determining red blood cell survival.

3.7 Labeling. The primary bag shall bear a label as specified in MIL-B-36939. In addition, the container lot or control number shall be shown. The secondary bag shall bear a label with the legend "Transfer Bag" or similar wording, and the container lot or control number. Except for the markings, the label for the secondary bag shall be as specified in MIL-B-36939. Labeling shall conform with the requirements of the Federal Food, Drug and Cosmetic Act. Shall be labeled in the metric system and the U.S. Customary System of Weights and Measures. When it is current industry practice to label in the metric system, the item is not required to be labeled in the U.S. Customary System of Weights and Measures.

3.8 Instruction for use. The unit package shall be permanently and legibly marked with complete instructions for use in the collecting, administration, and component processing of blood as they pertain to the particular set.

3.9 Workmanship. Workmanship shall be first class throughout. The blood collecting-dispensing bags (sets) shall be free from defects which detract from their appearance or may impair their serviceability.

3.10 Public Health Service Regulations. The equipment shall be in strict accordance with the Bureau of Biologics, Food and Drug Administration Code of Federal Regulation Part 273.

3.11 Compliance with the Federal Food, Drug, and Cosmetic Act. The items furnished under this specification must meet the requirements of the Federal Food, Drug, and Cosmetic Act. The contractor shall submit evidence to the purchasing agency prior to award of contract that the items he proposes to furnish under this specification meet the requirements of the Federal Food, Drug and Cosmetic Act, including an Effective New Drug Application (NDA) on

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file with the Food and Drug Administration.

3.12 Delivery. Not less than 44 months of the expiration dating period (potency period) shall remain at time of delivery to the Government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the plastic, the anticoagulant solution, or the steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Shall be in accordance with MIL-B-36939.

4.2.2 For pyrogen, sterility, and safety tests. Sampling procedures for pyrogen, sterility, and safety tests shall be conducted as indicated in the U.S.P., for transfusion and infusion assemblies.

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4.2.3 For in vivo tests. Each plastic batch shall be tested as specified in 4.4.6. If the material fails more than one test of the three (or four) tests required to be performed, the plastic batch represented shall be rejected.

4.2.4 For other tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table I.

Table I - Sampling Criteria for Tests

Component	Characteristics	Requirement	Test procedure	Unit of product	Insp. level	AQL (Percent Defective)
Primary and secondary bag	Security of attachment (connecting tube)	3.2.3.1	4.4.1	1-set	S-3	1.5
	Control clamp test	3.2.4	4.4.2	1-set	S-1	*
Primary bag	Vapor transmissivity	3.1	**	1-set	S-1	*
	Leakage test	3.2.1	**	1-set	S-4	1.5
	Infusion test	3.2.1	**	1-set	S-4	1.5
	Security of assembly:					
	Donor tubing to bag	3.2.1	**	1-set	S-3	1.5
	Donor tubing to needle hub	3.2.1	**	1-set	S-3	1.5
	Adhesion of label	3.2.1	**	1-set	S-3	2.5
	Bag suspension	3.2.1	**	1-set	S-3	1.5
	Security of attachment of cannula to hub	3.2.1	**	1-needle	S-3	1.5
	Corrosion-resistant stiffness, elasticity, and resistance to breakage of cannula	3.2.1	**	1-needle	S-1	*
	Freedom from foreign matter	3.2.1.1	4.3	1-needle	S-4	1.5
Secondary bag	Vapor transmissivity	3.2.2	**	1-set	S-1	*
	Leakage test	3.2.2	**	1-set	S-4	1.5
	Infusion test	3.2.2	**	1-set	S-4	1.5
	Adhesion of label	3.2.2	**	1-set	S-3	2.5
	Bag suspension	3.2.2	**	1-set	S-3	1.5

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Table I - Sampling Criteria for Tests (Cont'd)

Component	Characteristic	Requirement	Test procedure	Unit of product	Insp. level	AQL (Percent defective)
	Material - Class II	3.1	**	1-set	**	*
	Heavy metals	3.1	**	1-set	**	*
Set	Leaching	3.1	**	**	S-2	*
	Dimensions	3.2.1 and 3.2.2	--	1-set	S-4	1.5
	Fluid fill	3.2.1 and 3.2.2	**	1-set	S-4	1.5

*Acceptance number shall be zero for all sample sizes.
**See MIL-B-36939.

4.2.5 Raw materials. Four ounces of each lot of each metal used in the construction of the sets shall be selected for chemical analysis.

4.2.6 Components. Sufficient samples of the components shall be selected from each lot of material used in the manufacture of the sets to perform tests as required by the applicable subsidiary specifications or by tests specified herein. When sample size for components covered by the subsidiary specifications is not specified, the sample size for each test shall be as indicated above. No test result shall fall below the minimum or exceed the maximum values specified. The sample unit for testing shall be the quantity necessary to perform all required tests one time each. Not less than two samples shall be taken regardless of lot size.

4.2.7 Test samples. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The sets shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with MIL-STD-105 and the classification of defects specified in MIL-B-36939.

4.3.1 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of Section 5. Defects shall be scored but not limited to those as specified in Table II. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

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Table II. Examination of Preparation for Delivery

Examine	Defect
Containers	Not as specified.
Contents	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers. Cushioning inadequate, improper or missing.

4.3.1.1 Examination of closure and reinforcement for containers. When shipping containers are required to comply with PPP-B-636, examination for defects in closure and reinforcement shall be in accordance with the appendix to that specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements. Except as specified below, tests shall be conducted in accordance with MIL-B-36939.

4.4.1 Security of attachment (connecting tube). A force of 5 pounds shall be gradually applied, and maintained for 10 minutes, to the juncture of the connecting tube with the primary bag. This same test shall then be applied at the juncture with the secondary bag.

4.4.2 Control clamp performance test. The primary bag of an assembled set shall be filled with water (or saline solution). The set shall be suspended, in accordance with manufacturer's instructions (see 3.8), for component processing (plasma separation). A constant pressure of 2 pounds per square inch gage (p.s.i.g.) shall be applied to primary bag fluid. This pressure shall be maintained throughout the test. The ability of the clamp to accomplish complete shut-off, to regulate flow, and to maintain a constant rate of flow, shall be tested with the clamp at a point on the connecting tube midway between the two bags. The rate of flow shall be determined by recording the time required to collect a measured volume of fluid in the secondary bag.

4.4.3 Sterility tests. The method of testing whereby the liquid is drained from the bag through the donor tubing and needle satisfies the sterility test requirements for liquids and solids. Exterior of needle shall be flushed with anticoagulant solution.

4.4.3.1 Liquids. The anticoagulant solution shall be tested in accordance with the USP, for sterility tests for liquids.

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4.4.3.2 Solids. The solids shall be tested in accordance with the USP, for sterility tests for transfusion and infusion assemblies.

4.4.4 Safety test. The complete set shall be tested for safety in accordance with the method specified in the USP, for transfusion and infusion assemblies.

4.4.5 Pyrogen tests. The anticoagulant solution contained in the set shall be tested for freedom from pyrogens in accordance with the method specified in the USP. The method of testing whereby the liquid is drained from the bag through the donor tubing and needle satisfies the pyrogen test requirements.

4.4.6 In Vivo test. See Appendix. Three tests, as described in Appendix shall be performed, except when the material fails one test, a fourth test shall be performed. The tests shall be performed on finished equipment fabricated from, and representing, the plastic batch being tested.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One can containing three blood collecting-dispensing bags and donor sets, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Three blood collecting-dispensing bags and donor sets shall be packaged in a polyethylene bag having an average wall thickness of not less than 0.002 inch, with no wall thickness at any point below 0.00125 inch. The sets shall be packed in the bag in such a manner to prevent kinking of tubing and to retain sterility. Closure of the bag shall be effected by twisting the opening of the bag and securing. The bag containing the three sets shall be further packaged in a can conforming to type 1 of PPP-C-96 and fabricated of 1.25-pound hot dipped tinplate or 0.25-pound electrolytic tinplate. Interior coating shall conform to paragraph 3.2.1 of PPP-C-96. The can shall be provided with an aluminum pull top lid.

NOTE: The complete package (inner and outer containers) shall also conform to the following:

(a) Size. The completed package shall be of such a size as to utilize the minimum of storage or shipping space.

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(b) Treatment. The materials used for packaging shall be of such nature or treated to prevent growth of mold or spore bearing organisms. Chemical fungicides will not be permitted unless evidence can be submitted that they do not penetrate the bag.

(c) Shelf life. The sets being supplied shall be fabricated of such material and so packaged as to maintain units serviceable for a period of not less than four years when subjected to military depot storage conditions at temperatures below 86°F.

(d) Sealing. The completed package shall be hermetically sealed and shall be tamperproof and prevent opening or reclosure without destroying the seal.

5.1.2.2 Intermediate package. Eight unit packages (cans) shall be packaged in a snug fitting box constructed in accordance with PPP-B-636, class domestic. The box shall have a minimum bursting strength of 200 pounds. Closure of the box shall conform to method I as specified in PPP-B-636.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A. Two intermediate packages (16 cans) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant, grade V3c or V3s, style RSC. Closure, waterproofing and strapping shall be as specified in the appendix of the box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B. Two intermediate packages (16 cans) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class domestic and having a minimum bursting strength of 275 pounds. Closure shall conform to method I or II as specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, constructed so as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.2.4 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.3 Marking.

5.3.1 Unit package. Each can shall be lithographed with the information as specified for the unit package in MIL-STD-129. In addition, marking shall include the lot (control) number, contract number, name of contractor, requirements of statute, instructions for use (see 3.8), expiration date, and the method of sterilization (G-Gas, R-Radiation, S-Steam, C-Cold, D-Dry heat sterilization) shall be shown following the level of packaging and date of packaging: Example: "A-2/73-G". Packaging shall be on or after indicated packaging date. When the contractor is not the manufacturer, the name of the manufacturer shall also be shown. The lot (control) number, and contract number may be applied to the bottom lid of the can. In addition, marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F)".

5.3.2 Intermediate package. Intermediate package shall be marked in accordance with MIL-STD-129. Lot (control) number, contract or purchase order number and name of contractor shall be shown. In addition, the expiration date, the method of sterilization and the legend as specified in 5.3.1 shall also be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number, expiration date, the method of sterilization and the legend as specified in 5.3.1 shall also be shown.

6. NOTES

6.1 Intended use. The blood collecting component processing and dispensing system of connected bags and integral donor set is intended for collection, storage and processing of 450 ml of blood into its various components within a completely closed and tamperproof system.

6.2 Ordering data. Procurement document should specify the following:

(a) Number, title and date of this specification.
(b) Selection of applicable level of packing. (see 5.1 and 5.2)

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6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog:

National Stock No.	Item Identification
6515-00-079-9510	BLOOD COLLECTION-DISPENSING BAG AND DONOR SET, DOUBLE BAG, with 63 cc Anticoagulant, Disposable, 3s

It does not include all types or sizes as indicated by the title, nor all types which are commercially available.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1915
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APPENDIX

POST-TRANSFUSION SURVIVAL STUDIES OF STORED RED CELLS TAGGED WITH CR^{51}

PROCEDURE

Sodium Chromate Tagging

One-half ml. of sterile sodium chromate solution, as obtained from the Abbott Laboratories, Oak Ridge, Tennessee, containing approximately 500 microcuries of Cr^{51} as $Na_2Cr^{51}O4$, is diluted to a volume of 5 ml. with 0.9% sodium chloride solution.

Twenty-eight ml. of citrated stored blood are removed and mixed with a volume of the dilute sodium chromate containing approximately 60 microcuries. After gentle mixing, the blood is permitted to incubate in a 37° C. water bath for 20 minutes. We have found that a maximal amount of Cr^{51} is taken up by the erythrocytes within this period of time, but the uptake varies between 85 and 95% of the amount available, depending on the hematocrit of the blood, i.e., the higher the hematocrit the larger the volume of red cells and the higher the Cr^{51} uptake, provided the volume of the sample of whole citrated blood remains constant.

The specific activity of the sodium chromate must be sufficiently high so that the resulting concentration of chromium will not exceed 3 micrograms per ml. of packed red cells.

At the end of the period of incubation, the blood is gently but thoroughly mixed, and 10 ml. are withdrawn into a calibrated syringe and immediately and rapidly injected into the recipient. At the same time, a portion of the remaining chromated blood is centrifuged and the plasma is separated.

The uptake of Cr^{51} by the red cells is determined by the differential count on diluted samples of whole blood and plasma, as follows: a 1:100 dilution of the remaining chromated blood in distilled water and a 1:25 dilution of the separated plasma in saline solution are made.

Radioactivity measurements are carried out by counting gamma emission from a 3 ml. sample of the diluted material placed in a flat bottom test tube, using a well-type scintillation counter provided with a thallium activated sodium iodide crystal. The radioactivity is expressed as counts per minute per ml. cpm/ml of packed cells, after correction for dilution and subtraction of plasma activity.* To determine the survival rate of transfused cells, samples of 3 ml. of undiluted blood are used for radioactivity measurements; these samples of blood are centrifuged, after being placed in the flat bottom tubes, prior to counting.**

*All values are corrected for decay of radioactivity when indicated.
**One ml. of the plasma is diluted with 2 ml. of distilled water for purpose of counting and correction.

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Determination of Erythrocytes Survival

When the period of storage of injected blood is very short, such as one or two days, the survival of red cells at any time can be calculated from the following formula:

$$\frac{\text{cpm/ml. of packed cells at any time}}{\text{Immediate post-transfusion cpm/ml. of packed cells}}$$

The immediate post-transfusion count can be taken at any time between 3 and 20 minutes post-transfusion. Again it is emphasized that this simple formula is applicable only when relatively fresh blood is transfused.

To obtain sharp and reproducible results, the time of injection of the 10 ml. of tagged red cells should be as short as possible, between 7 and 10 seconds. Under these conditions, and when a normal subject is used as a recipient, the mixing time of the transfused red cells is 3 minutes or less.

In all other instances, i.e., when the red cells have been stored for a period of weeks, the survival of transfused cells is determined by the use of the 2 following formulae:

$$\text{100\% survival of transfused red cells} = \frac{\text{Total cpm of transfused red cells}}{\text{Total red cell vol. ml.}}$$

$$\text{Apparent \% survival at any time} = \frac{\text{cpm/ml. of whole blood} \times \frac{100}{\text{hematcrit}} \times 100}{\text{100\% survival}}$$

The blood volume determination can be carried out either by measuring directly the volume of red cells with the P^{32} method, or by the measure of the plasma volume with the Evans blue technique and the determination of the hematocrit, corrected by the factor of 0.91* to obtain the total body hematocrit.

In our routine studies, when stored blood is used, we carry out routinely determinations at 20 minutes after completion of the transfusion, and at 24 hours after transfusion. Thereafter, the counts are carried out at intervals of 1 or 2 days for 10 days. The average survival at 10 days of fresh, transfused red cells stored in glass bottles is 80% $\pm$ 4%.

*X 0.91

It is generally accepted that if the survival of transfused cells 24 hours after transfusion is 70% or better, preservation is considered satisfactory.

Details of Calculations: (based on counts per minute per milliliter)

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REMAINING CHROMATED BLOOD:

$$\frac{\text{cpm of diluted blood} \times 100}{3} = \text{cpm/ml. of blood}$$

$$\frac{\text{cpm of diluted plasma} \times 25}{3} = \text{cpm/ml. of plasma}$$

$$\text{cpm/ml. of plasma} \times \frac{(100 - \text{Hct})}{100} = \text{cpm/in plasma/ml. of blood}$$

cpm/ml. blood - cpm/in plasma/ml. blood = cpm/in cells/ml. of blood

cpm/in cells/ml. blood X 10 = ACTIVITY INJECTED ON THE CELLS

$$\frac{\text{cpm in cells/ml. of blood} \times 100}{\text{cpm/ml. of blood}} = \%\ \text{Uptake by the red cells}$$

POST INJECTION SAMPLES:

$$\text{cpm/ml. plasma} \times \frac{(100 - \text{Hct})}{100} = \text{cpm in plasma/ml. of blood}$$

$$\frac{\text{cpm in blood}}{3} - \text{cpm in plasma/ml. blood} = \text{cpm in cells/ml. of blood}$$

$$\frac{\text{cpm in cells/ml. of blood}}{\frac{\text{Hct}}{100}} = \text{cpm/ml. of red cells in sample}$$

References:

Max M. Strumia, M.D., Lawrence Taylor, Albert B. Sample, Louise S. Colwell, Ann Dugen: Uses and Limitations of Survival Studies of Erythrocytes Tagged with Cr^{51} Blood, Vol. X No. 5, 429, May, 1955.

John G. Gibson II, M.D.: A Method Employing Radioactive Chromium for Assaying the Viability of Human Erythrocytes Returned to the Circulation after Refrigerated Storage, J. Lab. & Clin. Med.

NOTE:

The hematocrit used in the determination referred to above is the one employing a capillary tube with very high centrifugal force, resulting in near absolute packing of red cells. Hence, no correction for trapped plasma need be made (see "An Improved Micro Hematocrit Method", by Max M. Strumia, et al, The American Journal of Clinical Pathology, Vol. 24, No. 9, September, 1954).

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DOCUMENT IDENTIFIER AND TITLE

MIL-B-37158 BLOOD COLLECTING DISPENSING BAG AND NEEDLE SET, Double Bag, With 63 cc Anticoagulant, Disposable, 3s

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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